we've discussed the "advantage actor-critic" part of the name at a conceptual level. Let's do the same for the "distributed" part now.

For virtually all deep learning models we do *batch training*, where a random subset of our training data is batched together and we compute the loss for this entire batch before we backpropagate and do gradient descent. This is necessary because the gradients, if we trained with single pieces of data at a time, would have too much variance, and the parameters would never converge on their optimal values. We need to average out the noise in a batch of data to get the real signal before updating the model parameters.

For example, if you're training an image classifier to recognize hand-drawn digits, and you train it with one image at a time, the algorithm would think that the background pixels are just as important as the digits in the foreground; it can only see the signal when averaged together with other images. The same concept applies in reinforcement learning, which is why we had to use an experience replay buffer with DQN.

Having a sufficiently large replay buffer requires a lot of memory, and in some cases a replay buffer is impractical. A replay buffer is possible when your reinforcement learning environment and agent algorithm follow the strict criteria of a Markov decision process, and in particular, the Markov property. Recall the Markov property says that the optimal action for a state S_t can be computed without reference to any prior states S_{t-1}; there is no need to keep a history of previously visited states. For simple games, this is the case, but for more complex environments, it may be necessary to remember the past in order to select the best option now.

Indeed, in many complex games it is common to use recurrent neural networks (RNNs) like a long short-term memory (LSTM) network or a gated recurrent unit (GRU). These RNNs can keep an internal state that can store traces of the past (figure 5.7). They are particularly useful for natural language processing (NLP) tasks where keeping track of preceding words or characters is critical to being able to encode or decode a sentence. Experience replay doesn't work with an RNN unless the replay buffer stores entire trajectories or full episodes, because the RNN is designed to process sequential data.